

ZHOU, CONGHAO

UC Santa Cruz
Physics Department
1156 High Street
Santa Cruz, CA 95064

CONTACT

email zhou.conghao@ucsc.edu
website <https://zch.info>

RESEARCH INTERESTS

dark matter · gravitational lensing · galaxy clusters · large scale structure
shear calibration · image simulation

EDUCATION

	University of California, Santa Cruz
Expected 2027	Ph.D. in Physics Physics Department Thesis: <i>TBD</i> Committee: TESLA E. JELTEMA, ALEXIE S. LEAUTHAUD
	Duke University
2020	B.A. in Physics with Highest Distinction Department of Physics Thesis: <i>Intrinsic Alignment of redMaPPer Clusters in the Dark Energy Survey</i> Committee: MICHAEL A. TROXEL, CHRISTOPHER W. WALTER, ROBERT L. WOLPERT

SCIENCE PUBLICATIONS WITH MAJOR CONTRIBUTIONS

[†] STUDENT ADVISED

1. No Cluster Left Behind I: Joint Mass Calibration for Dark Energy Survey and South Pole Telescope Clusters, in prep.
C. ZHOU, A.N. SALCEDO, H.-Y. WU, S. GRANDIS et al.
2. Response Bias in Shear Calibration, in prep.
C. ZHOU, M.R. BECKER, M. GATTI et al.
3. Cluster finding with outskirts stellar masses and percolation,
[arXiv:2608.19768 \(2026\)](#)
P. AVALOS[†], C. ZHOU, T. JELTEMA, K. LEIDIG, S. XU et al.
4. Accurate Galaxy Cluster Shear and Mass Calibration for LSST with AnaCal,
[arXiv:2606.21141 \(2026\)](#)
C. ZHOU, X. LI et al.
5. Relationship between 2D and 3D galaxy stellar mass and correlations with halo mass, *J. Cosmol. Astropart. Phys.* **10**, 014 (2025) [[arXiv:2502.05158](#)]
C. ZHOU, A. LEAUTHAUD, S. XU, B. DIEMER, S. HUANG et al.
6. Forecasting the constraints on optical selection bias and projection effects of galaxy cluster lensing with multiwavelength data, *Phys. Rev. D* **110**, 103508 (2024) [[arXiv:2312.11789](#)], *Editors' Suggestion*
C. ZHOU, H.-Y. WU, A.N. SALCEDO, S. GRANDIS, T. JELTEMA, A. LEAUTHAUD et al.
7. The intrinsic alignment of red galaxies in DES Y1 redMaPPer galaxy clusters, *Mon. Not. R. Astron. Soc.* **526**, 323 (2023) [[arXiv:2302.12325](#)]
C. ZHOU, A. TONG, M.A. TROXEL, J. BLAZEK, C. LIN et al.

TECHNICAL PUBLICATIONS WITH MAJOR CONTRIBUTIONS

1. Surface brightness profiles around massive galaxies in LSSTComCam data, [LSST Technote SITCOMTN-165 \(2025\)](#)
C. ZHOU, T. JELTEMA, A. VON DER LINDEN, L.S. KELVIN et al.

CONTRIBUTING AUTHOR SCIENCE PUBLICATIONS

1. The Connection between Halo Assembly History and the Stellar Distribution I: The Central Stellar Mass–Richness Plane, [arXiv:2609.14488 \(2026\)](#)
S. XU et al.
2. Probing Projection Effects in Optically-Selected Clusters with Velocity Dispersions Outliers, [arXiv:2609.08095 \(2026\)](#)
K. GRAHAM et al.
3. OASIS: Observation-Aware Simulation-Based Inference via Distributional Matching, [arXiv:2606.22572 \(2026\)](#)
A. FARAH et al.
4. The Vera C. Rubin Observatory Data Preview 1, [Astron. J. 171, 360 \(2026\)](#) [[arXiv:2603.23786](#)]
VERA C. RUBIN OBSERVATORY TEAM et al.
5. DESI-DR1 3×2 -pt analysis: consistent cosmology across weak lensing surveys, [Open J. Astrophys. 9 \(2026\)](#) [[arXiv:2512.15960](#)]
A. PORREDON et al.
6. NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1), [Astrophys. J. Lett. 1001, L35 \(2026\)](#) [[arXiv:2507.13409](#)]
C.O. CHANDLER et al.
7. Data Release 1 of the Dark Energy Spectroscopic Instrument, [Astron. J. 171, 285 \(2026\)](#) [[arXiv:2503.14745](#)]
DESI COLLABORATION et al.
8. Impact of projection-induced optical selection bias on the weak lensing mass calibration of galaxy clusters, [Phys. Rev. D 113, 063559 \(2026\)](#) [[arXiv:2510.00753](#)]
T.N. NDE et al.
9. Reaching for the Edge II: Stellar Halos out to Large Radii as a Tracer of Dark Matter Halo Mass, [arXiv:2511.10723 \(2025\)](#)
K. LEIDIG et al.
10. Cosmological Constraints from Dark Energy Survey Year 1 Cluster Lensing and Abundances with Simulation-based Forward-Modeling, [arXiv:2510.25706 \(2025\)](#)
A.N. SALCEDO et al.
11. Testing the Stellar Feedback-driven Breathing Mode in Low-mass Galaxies with Gas Kinematics, [arXiv:2510.17996 \(2025\)](#)
Y. LUO et al.
12. Dark energy survey year 3 results: Cosmological constraints from cluster abundances, weak lensing, and galaxy clustering, [Phys. Rev. D 112, 083535 \(2025\)](#) [[arXiv:2503.13632](#)]
T.M.C. ABBOTT et al.
13. The Outskirt Stellar Mass of Low-redshift Massive Galaxies Is an Excellent Halo Mass Proxy in Illustris/IllustrisTNG Simulations, [Astrophys. J. 990, 127 \(2025\)](#) [[arXiv:2412.03406](#)]
S. XU et al.
14. Association between optically identified galaxy clusters and the underlying dark matter halos, [Phys. Rev. D 112, 043517 \(2025\)](#) [[arXiv:2506.17526](#)]
S. CAO et al.
15. Improving galaxy cluster selection with the outskirts stellar mass of galaxies, [Phys. Rev. D 111, 123524 \(2025\)](#) [[arXiv:2410.20205](#)]
M. KWIECIEN et al.

16. Optical galaxy cluster mock catalogs with realistic projection effects: Validations with the SDSS clusters, *Phys. Rev. D* **111**, 063502 (2025) [arXiv:2410.02497]
A. LEE et al.
17. Dark Energy Survey Year 3 results: Simulation-based cosmological inference with wavelet harmonics, scattering transforms, and moments of weak lensing mass maps. Validation on simulations, *Phys. Rev. D* **109**, 063534 (2024) [arXiv:2310.17557]
M. GATTI et al.
18. Building an Efficient Cluster Cosmology Software Package for Modeling Cluster Counts and Lensing, arXiv:2309.06593 (2023)
M. AGUENA et al.
19. Dark Energy Survey Year 3 Results: Deep Field optical + near-infrared images and catalogue, *Mon. Not. R. Astron. Soc.* **509**, 3547 (2022) [arXiv:2012.12824]
W.G. HARTLEY et al.

TALKS

aug. 2026	Tsinghua University , Department of Astronomy Seminar <i>No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data</i>
jul. 2026	Tsung-Dao Lee Institute , Astronomy Seminar <i>No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data</i>
jul. 2026	Shanghai Jiao Tong University , Department of Astronomy Seminar <i>No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data</i>
may 2026	South Pole Telescope Collaboration , Cluster Working Group Meeting <i>No Cluster Left Behind: Joint Mass Calibration for DES and SPT Clusters</i>
dec. 2024	University of Arizona , Theoretical Astrophysics and Cosmology Seminar <i>Cluster Cosmology: Pixel to Object, CMB to Stellar Mass</i>
nov. 2024	University of California, Davis , Cosmology Seminar <i>Optical Cluster Cosmology: Selection to IA, Pixel to Object</i>
apr. 2024	CMB-S4 Collaboration , Cluster AWG Meeting <i>Calibrating Optical Selection Bias with Multiwavelength Data</i>
mar. 2024	Brown University , Center for the Fundamental Physics of the Universe Seminar <i>Cluster Cosmology Beyond Richness</i>
mar. 2024	University of Pennsylvania , Journal Club <i>Response Clustering Bias</i>
oct. 2023	Tsinghua University , Cosmology Seminar <i>Intrinsic Alignment in DES Y1 redMaPPer Clusters</i>
oct. 2023	Stony Brook University & Brookhaven National Lab , Cosmology Seminar <i>Constraining Optical Selection Bias in Cluster Lensing w/ Multiwavelength Data</i>

OBSERVING EXPERIENCE

10.5 effective nights	Dark Energy Spectroscopic Instrument, Mayall 4m at Kitt Peak
3 nights	DECaLS Intermediate Band Imaging Survey, Blanco 4m at CTIO

SCIENCE COLLABORATIONS

<i>2026–present</i>	Roman Science Collaboration
<i>2024–present</i>	Rubin Observatory Commissioning, In-Kind Contributor
<i>2023–present</i>	LSST Galaxies Science Collaboration
<i>2023–present</i>	LSST Informatics & Statistics Science Collaboration
<i>2020–present</i>	LSST Dark Energy Science Collaboration, Full Member since 2024
<i>2020–present</i>	Dark Energy Spectroscopic Instrument, Sponsored Member
<i>2018–present</i>	Dark Energy Survey, Student Member

PROFESSIONAL SOCIETIES

<i>2026–present</i>	Sigma Xi, Member
<i>2024–present</i>	American Astronomical Society, Student Member
<i>2024–present</i>	American Physical Society, Student Member

OUTREACH

<i>2023–2024</i>	Starlight Over Street Light
<i>2022</i>	UC Santa Cruz Physics Outreach Taskforce
<i>2022</i>	Science Internship Program Mentor
<i>2022</i>	UC Santa Cruz Physics Mentoring Program Mentor

COLLABORATION SERVICE

<i>jun. 2026–present</i>	LSST DESC Speaker's Bureau
<i>jul. 2022–present</i>	LSST DESC Mentoring Committee
<i>apr.2022–may.2025</i>	DES Early Career Scientists Committee

SEMINAR & MEETING ORGANIZATION

<i>2025 Winter</i>	UC Santa Cruz Physics Colloquium Committee
<i>sep.2023–jun.2024</i>	Organizer, CosmoGal Seminar, UC Santa Cruz
<i>jul.2023–jun.2024</i>	Organizer, BCG Seminar, Boise State
<i>jan. 2023</i>	LOC, APS Conference for Undergraduate Women in Physics at UC Santa Cruz
<i>aug. 2022</i>	SOC, LSST DESC 2022 Chicago Meeting

REFEREEING

<i>2024–present</i>	Astronomy & Astrophysics
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TEACHING EXPERIENCE

<i>2024 Spring</i>	Application of Quantum Materials (Grader)
<i>2024 Winter</i>	Electricity, Magnetism, and Optics (TA)
<i>2022 Fall</i>	Quantum Mechanics (TA)
<i>2022 Summer</i>	Intermediate Physics Laboratory (TA)
<i>2022 Spring</i>	Advanced Astrophysics Laboratory (TA)
<i>3 times</i>	Introductory Physics Laboratory (TA)

September 25, 2026